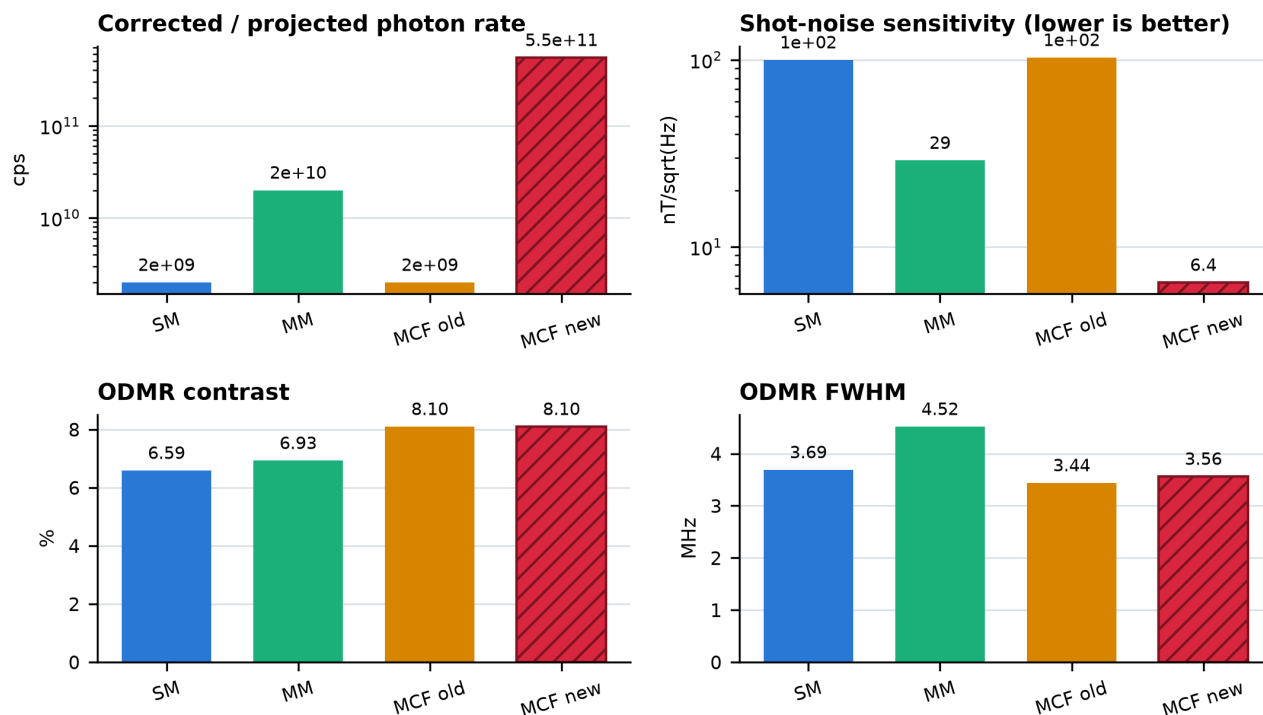


SM vs MM vs MCF optical probe comparison

Measured SM, MM and original MCF performance compared with the new monolithic seven-lens MCF model

Evidence boundary: SM, MM and MCF old are measured experimental values. MCF new is a shot-noise-limited, experiment-calibrated optical-model prediction and has not yet been experimentally validated.



Solid bars: measured experiment. Hatched bar: calibrated ideal-model prediction.

Primary comparison

Probe	Evidence	Photon rate*	Contrast	FWHM	Sensitivity
SM	Measured	2e+09 cps	6.59%	3.69 MHz	100.00 nT/sqrt(Hz)
MM	Measured	2e+10 cps	6.93%	4.52 MHz	29.00 nT/sqrt(Hz)
MCF old	Measured	2e+09 cps	8.10%	3.44 MHz	103.00 nT/sqrt(Hz)
MCF new	Calibrated model	5.48e+11 cps	8.10%	3.56 MHz	6.44 nT/sqrt(Hz)

* Measured rates are corrected for ND attenuation. The new-MCF rate is the experiment-calibrated optical model projection before an attenuation/detector chain.

Main result

Relative to the measured original MCF, the new design predicts 274.1x more detected photons and 16.0x lower shot-noise sensitivity, while retaining the measured 8.10% MCF contrast. The measured MM remains the best demonstrated sensitivity at 29 nT/sqrt(Hz); the new MCF value of 6.44 nT/sqrt(Hz) is a model target.

New MCF design and optical model

Printed structure	One monolithic IP-S body ($n = 1.52$): central lens plus six side lenses on a common support
Central channel	532 nm excitation; spherical surface; 20 μm aperture; 120 μm protrusion
Side channels	650-850 nm collection; six biconic surfaces; 24 μm aperture; 35 μm core pitch; 10 μm MFD
Best working gap	5.0 μm central-lens-to-diamond clearance for the fixed optimized tip
NV ensemble	3 ppm NV centers throughout the 80-90 μm layer in a 100 μm diamond
Optical physics	3-D rays, local surface normals, vector Snell law, Fresnel losses, wavelength-dependent diamond index, saturation and excitation-collection overlap

Complete monolithic MCF lens and 3-D Snell-law rays

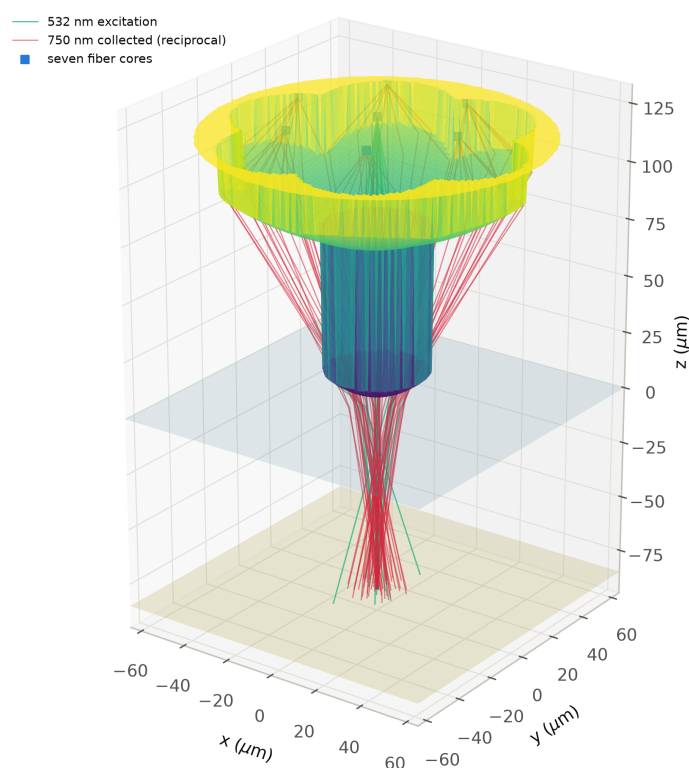


Figure 2. Complete seven-core printed-lens union. Green paths are 532 nm excitation rays. Red paths are reciprocal 750 nm collection rays. Refraction occurs only at the polymer-air and air-diamond boundaries.

Sensitivity model

The reported new-MCF sensitivity is calibrated to the measured MCF reference: $\eta_{\text{new}} = 103 \times (\text{FWHM}_{\text{new}} / 3.44) \times (0.081 / C_{\text{new}}) \times \sqrt{2e9 / R_{\text{new}}}$, with η in $\text{nT}/\sqrt{\text{Hz}}$ and R in counts/s .

Mounting tolerance and interpretation

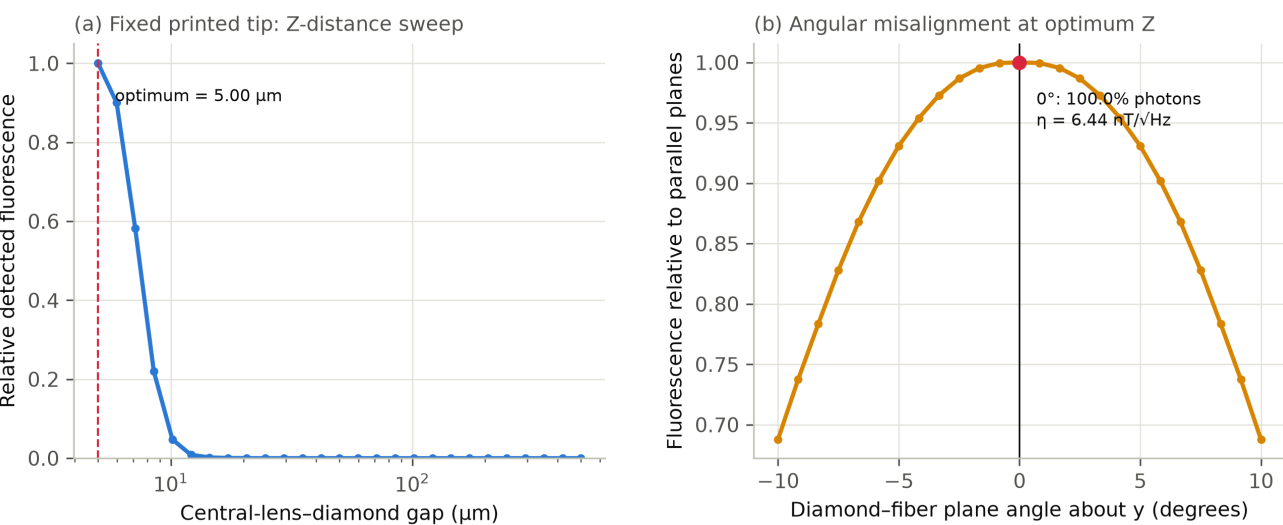


Figure 3. Fixed-lens Z-distance response and diamond-plane angular response. At nonzero tilt, all six side lenses are traced independently; the diamond and the 80-90 μm NV layer rotate together about the fiber y-axis.

Z sensitivity	The fixed lens is sharply optimized near 5 μm: relative fluorescence is 90.0% at 5.97 μm, 58.2% at 7.13 μm and 4.8% at 10.15 μm.
Angular sensitivity	At 1, 2, 5 and 10 degrees, predicted photon retention is 99.87%, 99.30%, 93.09% and 68.78%, respectively.
Measured ordering	SM and old MCF both give about 2e9 corrected cps. MM gives about 2e10 corrected cps and is therefore 10x higher in the supplied experiment.
New-design output	Projected 5.48e+11 cps, 8.10% contrast, 3.564 MHz FWHM, 6.44 nT/sqrt(Hz).

Limits of the comparison

The new-design photon rate is an ideal smooth-surface projection calibrated to the measured old-MCF result. It does not include detector saturation or dead time, fabrication roughness and shrinkage, alignment drift, real wavelength-dependent core modes, fluorescence background, filter transmission or detector quantum efficiency. No background-derived contrast penalty is applied because no measured background-versus-footprint calibration was supplied.

Accordingly, compare the new MCF with the measured probes as a design target and expected trend, not as a demonstrated experimental result. The next decisive test is to fabricate the full STL, measure the unattenuated linear photon range, and repeat contrast/FWHM/sensitivity measurements at controlled gap and tilt.

Reproducibility

Inputs: figures/mcf_freeform_design.json and figures/mcf_gap_angle_sweep.json. Geometry: figures/mcf_freeform_full_seven_core.stl. Regenerate design figures with 'python redesign_fig.py --angle-deg ANGLE'. Generate this report with 'python make_comparison_pdf.py'.